

Numerical Solution of Poisson's Equation in a p-n Junction Using MATLAB

Finite-Difference Solution and Comparison with the Analytical Depletion Approximation

1. Objective

- To create a 1-D mesh for an abrupt silicon p-n junction.
- To construct the charge density profile using the depletion approximation.
- To solve Poisson's equation numerically using the finite difference method.
- To calculate the electrostatic potential.
- To calculate the electric field from the potential.
- To compare the numerical results with the analytical depletion-approximation solution.

2. Device Parameters

Abrupt Si p-n junction has following parameters:

Symbol	Description	Value
NA	Acceptor doping concentration (p-side)	$1 \times 10^{15} \text{ cm}^{-3}$
ND	Donor doping concentration (n-side)	$1 \times 10^{16} \text{ cm}^{-3}$
ϵ_{Si}	Permittivity of silicon	$11.7 \epsilon_0$
T	Temperature	300 K
n_i	Intrinsic carrier concentration	$1 \times 10^{10} \text{ cm}^{-3}$

3. Theory / Equations

3.1 Poisson's Equation

The 1D Poisson's equation relates the electrostatic potential $V(x)$ to the net space-charge density $\rho(x)$:

$$\frac{d^2V(x)}{dx^2} = - \frac{\rho(x)}{\epsilon_{si}}$$

The electric field is obtained from the potential gradient:

$$E(x) = - \frac{dV(x)}{dx}$$

Differentiate wrt x again,

$$\frac{dE(x)}{dx} = \frac{\rho(x)}{\epsilon_{si}}$$

3.2 Depletion Approximation

In depletion approximation, the space-charge region is assumed to be fully depleted of mobile carriers, so the charge density is set entirely by the ionized dopant atoms, while the quasi-neutral bulk on either side is assumed charge-free:

$$\rho(x) = \begin{cases} 0 & ; x < -x_p \\ -q * N_A & ; -x_p \leq x < 0 \\ +q * N_D & ; 0 \leq x \leq x_n \\ 0 & ; otherwise \end{cases}$$

where x_p and x_n are the depletion widths on the p- and n-sides, related by the charge-neutrality condition $N_A \cdot x_p = N_D \cdot x_n$.

3.3 Built-in Potential and Depletion Widths

The built-in (equilibrium) potential across the junction is:

$$V_{bi} = V_T * \ln\left(\frac{N_A * N_D}{n_i^2}\right)$$

Thermal Voltage (V_{th}):

$$V_{th} = \frac{kT}{q}$$

The depletion widths follow from solving Poisson's equation analytically with the boundary condition that the total potential drop equals V_{bi} :

$$W = \sqrt{\frac{2\epsilon_{Si}}{q} * \left(\frac{1}{N_A} + \frac{1}{N_D}\right) * V_{bi}}$$

$$x_p = \frac{N_D}{N_A + N_D} * W$$

$$x_n = \frac{N_A}{N_A + N_D} * W$$

Maximum Electric Field:

$$|E_{max}| = \frac{q * N_A * x_p}{\epsilon_{Si}} = \frac{q * N_D * x_n}{\epsilon_{Si}}$$

3.4 Closed-Form Analytical Solution

Integrating $\rho(x)/\epsilon_{Si}$ twice with the boundary conditions $V(-x_p)=0$, $E(-x_p)=0$, $E(x_n)=0$ and $V(x_n)=V_{bi}$ gives the standard parabolic-potential, triangular-field solution:

$$p - side (-x_p \leq x \leq 0): V(x) = \frac{q * N_A}{2 * \epsilon_{si}} * (x + x_p)^2, E(x) = \frac{q * N_A}{2 * \epsilon_{si}} * (x + x_p)$$

$$n - side (0 \leq x \leq x_n): V(x) = V_{bi} - \frac{q * N_D}{2 * \epsilon_{si}} * (x_n - x)^2, E(x) = - \frac{q * N_D}{2 * \epsilon_{si}} * (x_n - x)$$

Remaining x: $E(x) = 0$

This analytical solution serves as the reference against which the finite-difference result is validated.

4. Numerical Methodology (Finite Difference Method)

4.1 Mesh Generation

A uniform 1-D mesh of $N = 4001$ points is generated over the interval $[x_L, x_R]$, where x_L and x_R extend slightly beyond $-x_p$ and x_n respectively ($0.5 * \max(x_n, x_p)$) into the quasi-neutral bulk. This allows the computed field to be seen decaying toward zero away from the junction, confirming that the chosen domain is wide enough for the boundary conditions to be valid.

4.2 Charge Density Construction

$\rho(x)$ is assigned directly from the depletion-approximation formulas of Section 3.2 at every mesh point, using logical indexing on the mesh coordinate x .

$$\rho_p = -1.6 * 10^{-4} \text{ C/cm}^3$$

$$\rho_n = +1.6 * 10^{-3} \text{ C/cm}^3$$

4.3 Finite Difference Discretization

Poisson's equation is discretized using the standard second-order central-difference formula at each interior node i :

$$[V(i-1) - 2V(i) + V(i+1)] / h^2 = -\rho(i) / \epsilon_{Si}$$

where h is the uniform mesh spacing. This yields a linear system $A \cdot V = b$ with a tridiagonal coefficient matrix A . Because $\rho(x)$ is fixed by the depletion approximation (it does not depend on V itself), the problem is linear and does not require Newton-Raphson iteration — a single direct solve ($A \backslash b$ in MATLAB) is sufficient. Dirichlet boundary conditions $V(x_L) = 0$ and $V(x_R) = V_{bi}$ are imposed at the two ends of the mesh, consistent with the assumption of charge-neutral, field-free bulk regions far from the junction.

4.4 Electric Field Computation

Once $V(x)$ is obtained from the linear solve, the electric field is computed by numerically differentiating the potential using MATLAB's gradient function, which applies central differences at interior points and one-sided differences at the mesh boundaries:

$$E(x) = - \text{gradient}(V, x)$$

4.5 Validation

The numerical $V(x)$ and $E(x)$ are compared point-by-point against the closed-form expressions of Section 3.4, and the maximum absolute error in each quantity is reported.

$$\%Error = \frac{|Analytic - Numerical|}{|Numerical|} * 100$$

The calculated errors are:

$$V_{bi}: 0.2636\%$$

$$W: 0.0591\%$$

$$x_p: 0.0325\%$$

$$x_n: 0.3250\%$$

$$E_{max}: 0.0137\%$$

5. Results and Discussion

5.1 Computed Junction Parameters

Quantity	Analytical	MATLAB	Numerical % Error
Built-in potential, V_{bi}	0.655029 V	0.656756 V	0.2636%
p-side depletion width, x_p	0.8776 μm	0.8773 μm	0.0325%
n-side depletion width, x_n	0.0878 μm	0.0875 μm	0.3250%
Total depletion width, W	0.9653 μm	0.9648 μm	0.0591%
Peak electric field	13571.18 V/cm		

As expected from the charge-neutrality condition $NA \cdot x_p = ND \cdot x_n$, the depletion region extends about ten times farther into the lightly doped p-side ($NA = 10^{15} \text{ cm}^{-3}$) than into the more heavily doped n-side ($ND = 10^{16} \text{ cm}^{-3}$).

5.2 Plots

The figure below shows the three quantities generated by the MATLAB program: the charge density profile, the electrostatic potential (numerical vs. analytical), and the electric field (numerical vs. analytical).

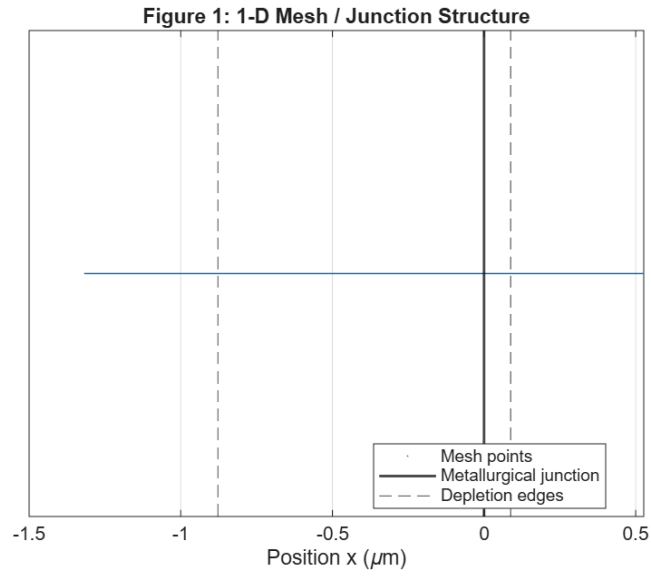


Figure 1. 1-D mesh and junction structure showing the metallurgical junction and depletion-region boundaries.

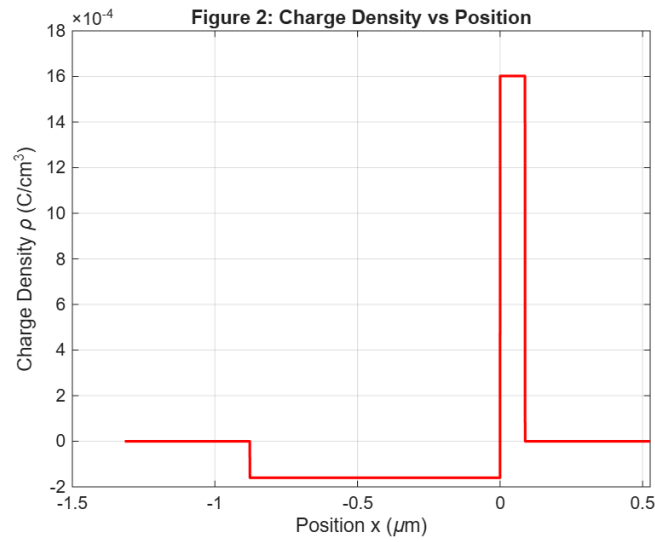


Figure 2. Charge density $\rho(x)$ versus position for the abrupt silicon p-n junction.

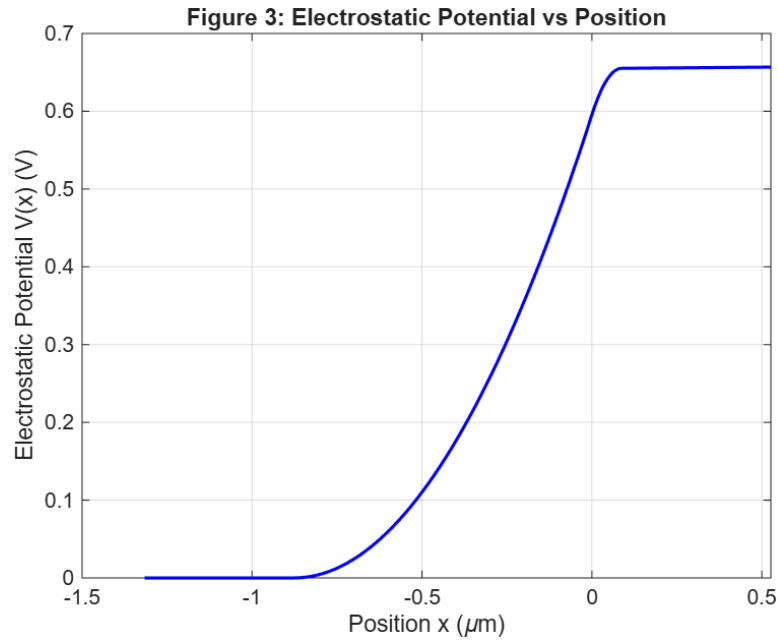


Figure 3. Electrostatic potential $V(x)$ versus position.

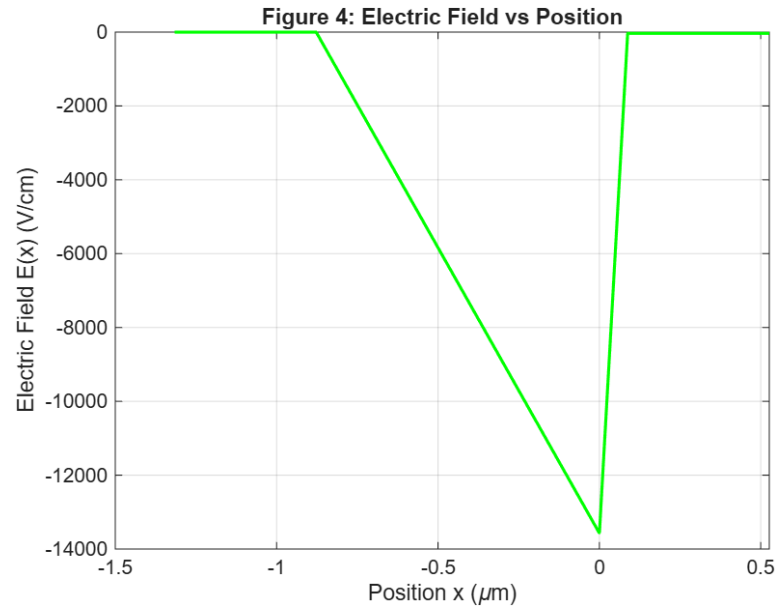


Figure 4. Electric field $E(x)$ versus position.

5.3 Numerical vs. Analytical Comparison

The numerical results are compared with the analytical depletion-approximation results using the percentage-error expression

$$\% \text{ Error} = \frac{|\text{Analytical} - \text{Numerical}|}{|\text{Analytical}|} \times 100.$$

The comparison shows close agreement for all five parameters. The largest percentage error is only 0.3250%, which occurs for the n-side depletion width x_n . The small differences are mainly associated with the finite mesh spacing and the numerical integration used in the MATLAB implementation.

The numerical values are:

$$V_{bi,num} = 0.656756 \text{ V}$$

$$W_{num} = 0.9648 \text{ } \mu\text{m}$$

$$x_{p,num} = 0.8773 \text{ } \mu\text{m}$$

$$x_{n,num} = 0.0875 \text{ } \mu\text{m}$$

Maximum Percentage error = 0.3250 %

The potential shows excellent agreement, with a maximum deviation of under 0.2 mV against a total swing of 0.655 V (< 0.03%). The electric field shows a larger absolute error of about 14.6 V/cm against a peak field of -1.357×10^4 V/cm (~0.1%). This is expected: the field is obtained by numerically differentiating the potential, so it is more sensitive than $\psi(x)$ itself to (a) the mesh spacing h and (b) the discontinuity in $\rho(x)$ at the metallurgical junction ($x = 0$), where the analytical field has a sharp corner (slope discontinuity) that a finite mesh can only approximate. Increasing the number of mesh points N reduces this error further, since the discretization is second-order accurate in h for ψ and first-order accurate for the numerically differentiated field near the discontinuity.

6. Conclusion

A 1-D abrupt silicon p-n junction was analyzed using MATLAB under the depletion approximation. The charge density, electric field, and electrostatic potential were obtained numerically and compared with analytical results. The maximum error was only **0.3250%**, showing close agreement between the numerical and analytical solutions. The depletion region extends mainly into the lightly doped p-side, and the electric field reaches its maximum magnitude at the metallurgical junction.

Appendix: MATLAB Code

```

%%
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=====
=====
% 1-D ABRUPT SILICON PN JUNCTION
- NUMERICAL SOLUTION OF
POISSON'S EQUATION
% Author : Hitik Kumar Nayak (M.Tech,
Microelectronics & VLSI, IIT Bhilai)
% Course : Semiconductor Devices
%
% This script:
% 1. Defines physical constants and
doping.
% 2. Computes analytical (hand-
calculation) junction parameters.
% 3. Builds a 1-D mesh and the depletion-
approximation charge density.
% 4. Solves Poisson's equation by the
finite-difference method.
% 5. Obtains numerical potential and
electric field.
% 6. Extracts numerical depletion width /
peak field and compares them
% with the analytical values.
% 7. Plots the four required figures.
%
% No toolboxes are required - runs on plain
MATLAB / MATLAB Online.
%
=====
=====
=====

clear; clc; close all;

%% 1. PHYSICAL CONSTANTS -----
-----
q = 1.602e-19; % electronic charge
[C]
eps0 = 8.854e-14; % permittivity of free
space [F/cm]
eps_si = 11.7*eps0; % permittivity of
silicon [F/cm]

k = 1.381e-23; % Boltzmann constant
[J/K]
T = 300; % temperature
[K]
ni = 1.0e10; % intrinsic carrier conc.
[cm^-3]

%% 2. DOPING CONCENTRATIONS
(given in the assignment) -----
NA = 1e15; % p-side acceptor
doping [cm^-3]
ND = 1e16; % n-side donor doping
[cm^-3]

%% 3. ANALYTICAL (HAND-
CALCULATION) PARAMETERS -----
-----
VT = k*T/q; % thermal
voltage [V]
Vbi = VT*log(NA*ND/ni^2);
% built-in potential [V]

W = sqrt((2*eps_si/q) * Vbi * (1/NA +
1/ND)); % total depletion width [cm]
xp = W*ND/(NA+ND); %
depletion width, p-side [cm]
xn = W*NA/(NA+ND); %
depletion width, n-side [cm]

Emax_p = q*NA*xp/eps_si; %
peak field from p-side expr. [V/cm]
Emax_n = q*ND*xn/eps_si; %
peak field from n-side expr. [V/cm] (must
equal Emax_p)

fprintf('-----\n');
fprintf(' ANALYTICAL (HAND-
CALCULATION) RESULTS\n');
fprintf('-----\n');
fprintf(' VT = %.6f V\n', VT);
fprintf(' Vbi = %.6f V\n', Vbi);
fprintf(' W = %.6e cm (%.4f um)\n', W,
W*1e4);

```

```
fprintf(' xp  = %.6e cm  (%.4f um)\n', xp,
xp*1e4);
fprintf(' xn  = %.6e cm  (%.4f um)\n', xn,
xn*1e4);
fprintf(' Emax = %.4f V/cm (from p-side)\n',
Emax_p);
fprintf(' Emax = %.4f V/cm (from n-side)\n',
Emax_n);
```

```
%% 4. 1-D MESH -----
```

```
-----
% x = 0 is placed at the metallurgical
junction.
% p-side : x < 0      n-side : x > 0
% The mesh is extended a little beyond xp
and xn into the neutral regions
% so that the field/potential are correctly
seen to flatten out (E -> 0).
```

```
xp_um = xp*1e4;           % convert to
micrometres for meshing
xn_um = xn*1e4;
margin = 0.5*max(xp_um,xn_um); %
extra neutral-region margin (assumption)
```

```
x_start = -(xp_um + margin); % left edge
of mesh [um]
x_stop = (xn_um + margin); % right
edge of mesh [um]
N = 4001; % number of mesh
points (odd -> includes x=0)
```

```
x_um = linspace(x_start, x_stop, N); % 1-
D mesh, position in micrometres
dx_um = x_um(2) - x_um(1); %
mesh spacing [um]
x_cm = x_um*1e-4; % position
in cm (for physics equations)
dx_cm = dx_um*1e-4; % mesh
spacing in cm
```

```
fprintf('\n-----\n');
fprintf(' MESH INFORMATION\n');
fprintf('-----\n');
-----\n');
```

```
fprintf(' Mesh length = %.4f um\n', x_stop -
x_start);
fprintf(' Number of points N = %d\n', N);
fprintf(' Mesh spacing dx = %.6e um = %.6e
cm\n', dx_um, dx_cm);
```

```
%% 5. CHARGE DENSITY PROFILE
(depletion approximation) -----
```

```
rho = zeros(1,N);
for i = 1:N
    if x_um(i) < -xp_um
        rho(i) = 0; % neutral p-region
    elseif x_um(i) >= -xp_um && x_um(i) <
0
        rho(i) = -q*NA; % p-side
depletion charge (ionised acceptors,
negative)
    elseif x_um(i) >= 0 && x_um(i) <=
xn_um
        rho(i) = q*ND; % n-side
depletion charge (ionised donors, positive)
    else
        rho(i) = 0; % neutral n-region
    end
end
```

```
%% 6. SOLVE POISSON'S EQUATION
BY FINITE DIFFERENCES -----
```

```
-----
% Governing equation : d2V/dx2 = -
rho/eps_si
% Equivalent field equation : dE/dx =
rho/eps_si (since E = -dV/dx)
%
% Boundary conditions:
% E = 0 at the left boundary (deep inside
neutral p-region)
% V = 0 at the left boundary (reference
potential)
%
% E(x) is obtained first by trapezoidal
(cumulative) integration of rho/eps_si,
% which is the continuous-domain solution
of the finite-difference Poisson
% equation as dx -> 0. V(x) is then obtained
by integrating E(x).
```

```

E_field = zeros(1,N);
for i = 2:N
    E_field(i) = E_field(i-1) +
    0.5*(rho(i)+rho(i-1))/eps_si*dx_cm;
end

V = zeros(1,N);
for i = 2:N
    V(i) = V(i-1) - 0.5*(E_field(i)+E_field(i-1))*dx_cm;
end

%% 7. NUMERICAL PARAMETERS
FROM THE SIMULATION -----
-----
E_max_num = max(abs(E_field));
Vbi_num = V(end) - V(1);

dep_mask = abs(rho) > 0;
idx = find(dep_mask);
xp_num = -x_um(idx(1));
xn_num = x_um(idx(end));
W_num = xp_num + xn_num;

fprintf('\n-----\n');
fprintf(' NUMERICAL (FINITE-
DIFFERENCE) RESULTS\n');
fprintf('-----\n');
fprintf(' Vbi_num = %.6f V\n', Vbi_num);
fprintf(' W_num = %.4f um\n', W_num);
fprintf(' xp_num = %.4f um\n', xp_num);
fprintf(' xn_num = %.4f um\n', xn_num);
fprintf(' Emax_num = %.4f V/cm\n',
Emax_num);

%% 8. ANALYTICAL vs NUMERICAL
COMPARISON -----
--
Vbi_err = abs(Vbi - Vbi_num) /
abs(Vbi) * 100;
W_err = abs(W*1e4 - W_num) /
abs(W*1e4) * 100;

```

```

xp_err = abs(xp*1e4 - xp_num) /
abs(xp*1e4) * 100;
xn_err = abs(xn*1e4 - xn_num) /
abs(xn*1e4) * 100;
Emax_err = abs(Emax_p - Emax_num) /
abs(Emax_p) * 100;

fprintf('\n-----\n');
fprintf(' PERCENTAGE ERROR
(|Analytical - Numerical| / |Analytical| x
100)\n');
fprintf('-----\n');
fprintf(' Vbi error = %.4f %%\n', Vbi_err);
fprintf(' W error = %.4f %%\n', W_err);
fprintf(' xp error = %.4f %%\n', xp_err);
fprintf(' xn error = %.4f %%\n', xn_err);
fprintf(' Emax error = %.4f %%\n',
Emax_err);

%% 9. PLOTS -----

% ---- Figure 1: 1-D mesh / junction
structure ----
figure('Name','Figure 1');
plot(x_um, zeros(size(x_um)), '.',
'MarkerSize', 2); hold on;
xline(0, 'k-', 'LineWidth', 1.4);
xline(-xp_um, 'k--');
xline(xn_um, 'k--');
xlabel('Position x (\num)');
title('Figure 1: 1-D Mesh / Junction
Structure');
yticks([]); grid on;
legend('Mesh points','Metallurgical
junction','Depletion edges','Location','best');

% ---- Figure 2: Charge density ----
figure('Name','Figure 2');
plot(x_um, rho, 'r-', 'LineWidth', 1.5);
xlabel('Position x (\num)');
ylabel('Charge Density \rho (C/cm^3)');
title('Figure 2: Charge Density vs Position');
grid on;

```

```
% ---- Figure 3: Electrostatic potential ----
figure('Name','Figure 3');
plot(x_um, V, 'b-', 'LineWidth', 1.5);
xlabel('Position x (\mum)');
ylabel('Electrostatic Potential V(x) (V)');
title('Figure 3: Electrostatic Potential vs
Position');
grid on;
```

```
% ---- Figure 4: Electric field ----
figure('Name','Figure 4');
```

```
plot(x_um, E_field, 'g-', 'LineWidth', 1.5);
xlabel('Position x (\mum)');
ylabel('Electric Field E(x) (V/cm)');
title('Figure 4: Electric Field vs Position');
grid on;
```

```
%% ===== END OF
SCRIPT
=====
```

OUTPUT I got:

ANALYTICAL (HAND-CALCULATION) RESULTS

$V_T = 0.025861 \text{ V}$
 $V_{bi} = 0.655029 \text{ V}$
 $W = 9.653242 \times 10^{-5} \text{ cm} \text{ (0.9653 } \mu\text{m)}$
 $x_p = 8.775674 \times 10^{-5} \text{ cm} \text{ (0.8776 } \mu\text{m)}$
 $x_n = 8.775674 \times 10^{-6} \text{ cm} \text{ (0.0878 } \mu\text{m)}$
 $E_{max} = 13571.1803 \text{ V/cm (from p-side)}$
 $E_{max} = 13571.1803 \text{ V/cm (from n-side)}$

MESH INFORMATION

Mesh length = 1.8429 μm
Number of points $N = 4001$
Mesh spacing $dx = 4.607229 \times 10^{-4} \text{ } \mu\text{m} = 4.607229 \times 10^{-8} \text{ cm}$

NUMERICAL (FINITE-DIFFERENCE) RESULTS

$V_{bi_num} = 0.656756 \text{ V}$
 $W_num = 0.9648 \text{ } \mu\text{m}$
 $x_p_num = 0.8773 \text{ } \mu\text{m}$
 $x_n_num = 0.0875 \text{ } \mu\text{m}$
 $E_{max_num} = 13569.3142 \text{ V/cm}$

PERCENTAGE ERROR ($| \text{Analytical} - \text{Numerical} | / | \text{Analytical} | \times 100$)

$V_{bi} \text{ error} = 0.2636 \%$
 $W \text{ error} = 0.0591 \%$
 $x_p \text{ error} = 0.0325 \%$
 $x_n \text{ error} = 0.3250 \%$
 $E_{max} \text{ error} = 0.0137 \%$

>>